

ANNETTE ISLAND, AK, USA

WMO: 703980

Lat: 55.039N

Lon: 131.579W

Elev: 109

StdP: 14.64

Time Zone: -9.00 (NAK)

Period: 96-19

WBAN: 25308

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	18.3	22.2	-1.5	5.1	23.5	4.0	6.8	26.8	31.8	44.2	28.2	44.3	5.7	60	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.6	75.1	61.9	70.9	60.4	67.3	58.5	63.5	72.8	61.6	68.9	59.8	65.4	6.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.5	76.3	65.2	58.3	73.0	63.4	57.3	70.3	61.8	28.8	73.0	27.3	68.8	26.2	65.4	70.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.7	20.6	18.3	DB	12.4	82.7	4.9	4.4	8.9	85.9	6.0	88.4	3.2	90.9	-0.3	94.0
			WB	11.2	66.3	4.8	2.1	7.7	67.8	4.9	69.1	2.1	70.2	-1.4	71.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	37.5	37.3	39.1	44.2	50.6	55.5	58.7	59.2	54.1	47.3	40.6	37.0
	DBStd	9.72	7.36	5.63	5.14	4.24	4.98	4.57	4.21	4.02	3.62	4.36	5.75	6.53
	HDD50	2114	388	355	337	184	52	2	0	0	8	102	283	403
	HDD65	6656	853	775	801	625	447	290	203	187	328	548	731	868
	CDD50	954	0	0	0	9	70	167	271	286	130	19	2	0
	CDD65	22	0	0	0	0	0	5	9	8	0	0	0	0
	CDH74	178	0	0	0	0	5	57	69	45	2	0	0	0
	CDH80	31	0	0	0	0	0	15	13	3	0	0	0	0
	WSAvg	7.3	8.6	7.6	8.0	7.5	6.6	6.3	5.8	5.9	6.5	7.7	8.1	8.5
(15) Precipitation	PrecAvg	104.1	10.4	8.4	8.2	7.6	5.9	4.8	4.8	6.8	9.9	14.4	11.7	11.1
	PrecMax	144.2	19.3	18.1	23.6	16.9	14.7	9.4	9.1	14.7	18.1	27.2	24.9	23.3
	PrecMin	77.7	3.0	0.1	2.1	1.9	0.5	0.6	0.6	0.8	2.5	4.6	3.6	3.4
	PrecStd	13.5	3.5	4.1	3.8	3.3	3.0	2.0	2.2	3.3	3.7	4.8	4.4	4.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.2	52.5	56.5	65.4	73.7	81.7	81.8	79.6	70.4	60.7	56.0	51.9
		MCWB	43.1	42.2	46.3	49.6	56.0	62.5	66.4	64.6	60.2	52.3	46.9	43.4
	2%	DB	49.3	49.2	50.7	58.4	67.7	72.5	74.4	73.7	65.2	57.3	52.2	49.5
		MCWB	45.5	42.5	43.6	46.9	53.9	60.0	63.2	62.7	57.3	51.0	47.5	45.0
	5%	DB	47.5	46.9	48.1	54.0	62.7	67.0	69.5	69.7	62.1	55.1	50.0	47.5
		MCWB	44.6	42.3	43.3	45.7	52.8	57.1	60.6	60.8	55.9	50.5	46.7	44.5
	10%	DB	46.1	45.1	46.2	50.7	58.8	62.8	65.6	66.3	59.5	53.4	48.1	45.8
		MCWB	43.5	42.1	42.3	44.5	51.0	55.6	58.7	59.4	54.9	50.0	45.4	43.4
	0.4%	WB	48.4	46.5	48.1	52.6	58.9	64.1	67.0	65.7	61.5	56.1	51.0	48.2
		MCDB	49.2	48.3	51.6	62.0	69.8	77.5	79.7	76.8	69.2	58.7	52.2	49.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.4	44.6	46.1	48.9	55.6	60.4	63.9	63.4	58.9	53.8	48.9	46.4
		MCDB	47.7	46.3	49.2	55.0	63.9	70.1	73.4	72.1	63.1	54.9	50.9	48.1
	5%	WB	45.6	43.7	44.4	46.9	53.6	58.4	61.4	61.7	57.3	52.2	47.4	45.2
		MCDB	47.0	45.7	47.2	52.0	59.9	65.6	68.3	68.6	60.2	53.5	49.4	46.9
	10%	WB	44.1	42.3	43.2	45.7	51.9	56.5	59.4	60.0	56.0	50.8	46.0	43.5
		MCDB	45.7	44.7	45.7	49.9	57.0	61.7	64.3	65.1	58.5	52.5	47.9	45.3
	5% DB	MDBR	8.1	9.5	9.7	10.8	11.5	10.9	10.3	10.6	9.8	9.0	8.6	8.0
		MCDBR	8.1	10.5	12.3	16.7	19.7	20.3	18.8	17.9	14.8	11.3	9.2	7.7
		MCWBR	5.8	6.8	7.6	8.4	9.5	9.2	8.9	8.6	8.4	7.2	6.8	5.7
		MCWBR	6.4	8.1	10.5	13.5	16.5	17.5	17.4	16.4	12.5	7.9	7.8	6.9
(38) Mean Daily Temperature Range	5% WB	MCWBR	5.8	6.5	7.4	7.7	8.9	8.4	8.5	8.3	7.7	6.4	6.7	6.1
(40) Clear-Sky Solar Irradiance	taub		0.300	0.324	0.334	0.366	0.363	0.367	0.373	0.359	0.345	0.335	0.308	0.293
	taud		2.392	2.413	2.450	2.367	2.442	2.462	2.479	2.543	2.557	2.478	2.356	2.292
	Ebn at Noon		205	242	268	272	280	279	275	273	262	234	196	174
	Edh at Noon		16	23	27	34	33	33	32	28	24	21	16	14
(44) All-Sky Solar Radiation	RadAvg		152	403	715	1174	1482	1548	1382	1207	788	431	194	119
	RadStd		20	79	114	88	149	120	174	117	102	68	32	22

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page